Gauthier Bonhomme

Final-year engineering student in **Mechatronics & Control** — Arts et Métiers (ENSAM), France

ESA student internship 2027 • 6 months, Feb – Jul 2027

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- **Hands-on builder** — CNC machining, sand casting, engine rebuild, metalwork.
- **Mechatronics & control** — robotics, PID/RST, FOC drives, microcontrollers.
- **Simulate, then validate** — FEA within ≈1 % of ASTM; CAM run on a turn-mill.

→ **Reports, workshop photos & live demos: gauthierbonhomme.com/esa**

Portfolio

Education

Arts et Métiers (ENSAM) — *Diplôme d'Ingénieur (MSc), Mechatronics & Control* 2024 – 2027
Robotics, control, drives, mechanical design, FEA, manufacturing — enrolled until Jul 2027.

Classe préparatoire PT, Lycée Mermoz, Montpellier — *maths, physics, engineering science* 2021 – 2024

Projects

CNC-machined engine crankshaft report ↗ 2025
Fusion 360 CAM → DMG MORI turn-mill; solved chatter and axis-limit issues on the machine.

Cast-iron foundry part — team of 4 report ↗ 2025 – 2026
QuickCast simulation, riser and gating design, hand-made sand moulds and cores, cast-iron pour.

Fracture-mechanics FEA (Abaqus) report ↗
Stress-intensity factor within ≈1 % of the ASTM E399-90 analytical value.

Workshop & prototyping — personal photos ↗
Motorcycle and car engines (one fully reconditioned), sheet-metal tables, Arduino builds.

Experience

Royal Canin — summer intern, plant maintenance software Jun – Aug 2026
Improved a pet-food plant's maintenance software, working daily with the maintenance teams.

Hall b — founding team, fitness startup Jun – Aug 2025
Launched the facility from scratch; built its site-selection model and real-time video analytics.

ÉPOPÉE GADZ — treasurer, Europ'Raid 2026 2025 – 2026
8,000-km charity raid across Europe: budget and sponsorship, nearly 4,000 euros raised.

Skills

Design & manufacturing: CATIA V5, Fusion 360 (CAM), CNC turn-milling, sand casting, 3D printing
Simulation & control: Abaqus, QuickCast, PID/RST synthesis, motor control (FOC), MATLAB (basic)
Programming & embedded: Python, Arduino/ESP32, Git, Linux
Languages: French (native), English (C1)

Interests

Ultra-trail (100-km finisher, preparing the UTMB) • alpinism • bouldering 7B • judo black belt